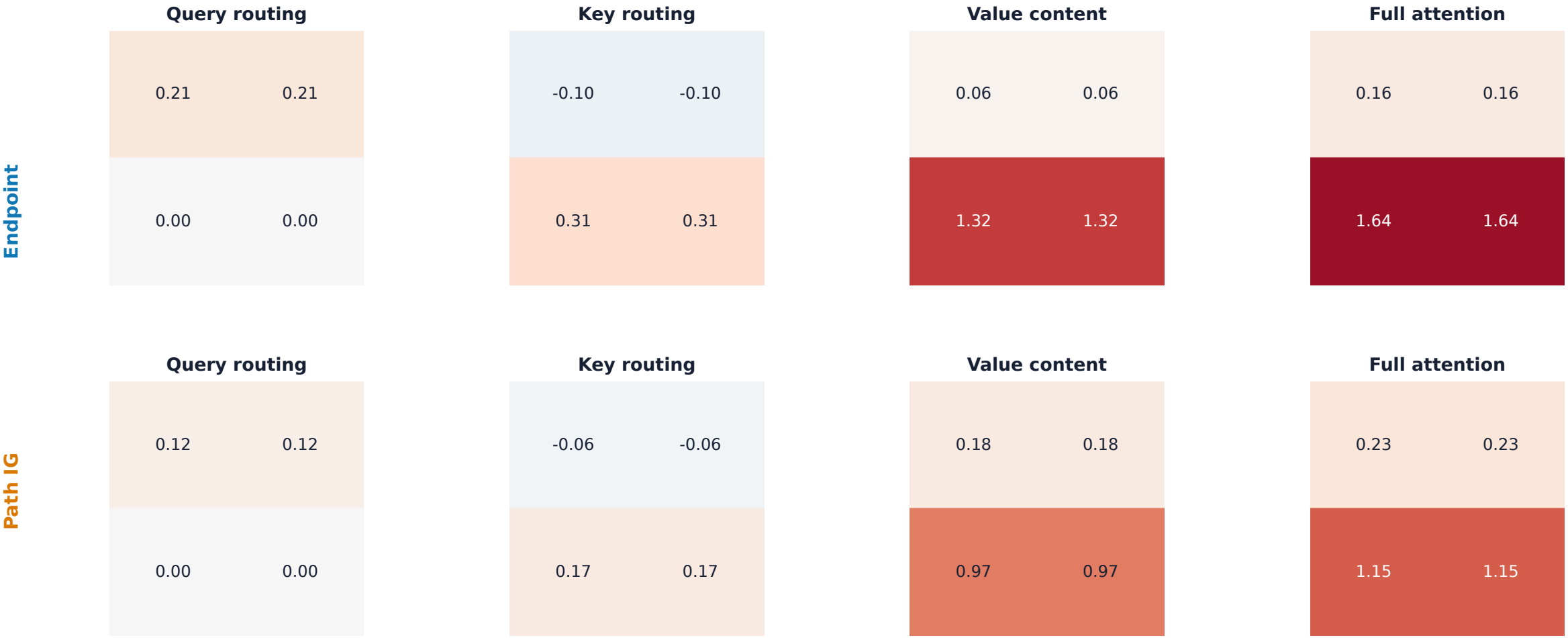
Attention separates into Query, Key, and Value routes

$$dO = J_{softmax}(S)[(dQ)K^T + Q(dK)^T]V + A(dV)$$



Within each row: Query + Key + Value = full attention. Freezing A keeps only a conditional Value-path target.